

VANELLE AYONTA NDJOUTSE

Data Scientist • AI & ML Engineer • NLP Engineer

Statistician-Economist Engineer (ISE) – Intelligent decision-support systems

Lieu : Yaoundé, Cameroun Tél. : +237 699 091 775 Email : vanelleayonta@gmail.com

LinkedIn : linkedin.com/in/vanelle-ayonta GitHub : github.com/vanelle-ayonta Web : vanelle-ayonta.github.io

PROFILE

Statistician-Economist Engineer (ISSEA) and **AI / Data Engineer** building **intelligent decision-support systems** end to end. I combine rigorous statistical and econometric expertise with strong command of **Machine Learning, Deep Learning** and **LLMs** to turn complex data – textual and numerical alike – into reliable decision levers. Experience gained inside a **central bank** (BEAC) and the **National Institute of Statistics**, where I shipped robust, secure and production-ready **agentic RAG, NLP** and document-automation pipelines. Autonomous and impact-driven, I transform raw data into clear, actionable insight.

PROFESSIONAL EXPERIENCE

AI & Data Science Engineer (Internship) Mar. 2026 – May 2026

Yaoundé, CM

BEAC – Bank of Central African States • Financial Inclusion & Innovation Dept.

- › Designed and deployed a complete **agentic RAG pipeline** – scraping of **22 international sources**, optimised chunking, **768D** embeddings, **FAISS** vector index, LLM agents and automated export (Word / PDF / e-mail) – fully automating the department's information watch.
- › Orchestrated a **multi-LLM** architecture with **Pydantic-AI**, secured by **guardrails** and **prompt-injection** protection.
- › Built a **10-page Streamlit** interface and Dockerised the solution for reproducible, production-ready deployment.
- › Automated extraction from **scanned PDFs** via **Claude API**, eliminating a time-consuming manual process.
- › Contributed to the redesign of the survey questionnaire and specifications of the **CEMAC** financial-inclusion data-collection platform.

Data Analyst (Internship) Aug. 2025 – Oct. 2025

Yaoundé, CM

INS – National Institute of Statistics • Business Statistics Dept.

- › Designed **statistical analysis plans** for national data-collection operations.
- › Processed and analysed structured survey databases and mined unstructured sources (reports, official documents) to produce **decision-ready syntheses**.

Mathematics Teacher & Department Head Nov. 2018 – Oct. 2023

Melong, CM

Melong Bilingual High School

- › Coordinated the mathematics department's teaching activities and **led a team of teachers**.
- › Drove learning assessment and academic planning over **5 years**.

PROJECTS & RESEARCH

NLP Analysis of Cameroon's Finance Laws 2026

Measure the semantic alignment of the 2026 Finance Law with the National Development Strategy (SND 30) objectives and assess budget compliance.

Stack : Python · sentence-transformers · CamemBERT · semantic similarity

→ Compliance-analysis pipeline over a legislative corpus, informing public-policy evaluation.

Deep-Learning Quality Control

2025–2026

Build a binary quality classifier from **28 features** and **2,148 samples**.

Stack : PyTorch · MLP · Optuna · Streamlit

→ Model tuned via hyperparameter search (Optuna) with an **interactive prediction dashboard**.

Geomarketing & Commercial Strategy

Optimise market entry and territorial coverage across banking, telecoms, mobile money, insurance and agribusiness in Cameroon.

Stack : Spatial analysis · segmentation · data visualization

→ Recommendations on bank-branch placement, telecom sales-force sizing and agribusiness territorial defence.

Data Collection & Delivery Automation

Make field collection and data access more reliable through custom web tools.

Stack : HTML · Google Sheets · Netlify · Claude API · Streamlit

→ Deployed a custom web questionnaire and an app for automatic data extraction from scanned PDFs.

TECHNICAL SKILLS

Programming	Python · R · SQL · Bash
Machine Learning	scikit-learn · SVM · predictive modelling · supervised & unsupervised learning
Deep Learning	PyTorch · neural networks (MLP) · Optuna
LLM & GenAI	Agentic RAG · Pydantic-AI · Claude API (Anthropic) · guardrails · prompt-injection defence
NLP	CamemBERT · sentence-transformers · Hugging Face · FAISS · semantic similarity
MLOps & Deployment	Docker · Git / GitHub · GitHub Actions · Streamlit · Netlify
Data & Visualization	Power BI · Metabase · Pandas · data storytelling
Databases	MongoDB · MySQL
Statistics & Econ.	Stata · SPSS · econometrics · sampling design
Data Collection	Survey Solutions · KoboToolbox · CPro · Google Forms
Soft skills	Analytical rigour · autonomy · leadership · communication to non-technical audiences

EDUCATION

Statistician-Economist Engineer Degree (ISE)	2023 – 2026
ISSEA — Sub-regional Institute of Statistics and Applied Economics	Yaoundé, CM
DIPES I — Mathematics (Teaching Diploma)	2015 – 2018
ENS — Higher Teacher Training College of Yaoundé	Yaoundé, CM
B.Sc. in Applied Mathematics	2015 – 2019
University of Yaoundé I	Yaoundé, CM

LEADERSHIP & ENGAGEMENT

Vice-President, Student Council

Apr. 2025 – Apr. 2026

- › Led student projects and mediated between the student body and the administration.

Cluster Leader – Teacher Training Program

Oct. 2019 – Oct. 2023

AIMS – African Institute for Mathematical Sciences

Melong, CM

- › Facilitated training groups, mentored peers and coordinated scientific communication.

CERTIFICATIONS & TRAINING

- › Monitoring & Evaluation of a development project.
- › Designing projects respectful of the environment and local populations.
- › Decentralisation: challenges and processes.

LANGUAGES

French

Native

English

Professional working (B2)

References available upon request.